Aileron Trim Tab Position Transmitter – Maintenance Practices

1. Description

A linear potentiometer is mounted on the outboard, port side of the aileron at approximately station 571.978 and sends a signal of the aileron trim position through the DAU and onto the Flight Data Recorder (if installed) and the aileron trim position is indicated on the system screen.

2. Removal/Installation

A. Remove Aileron Trim Tab Position Indicator Potentiometer

1) With aircraft power OFF, remove trim tab actuator assembly (refer to 27-13-11, Aileron Trim Tab Actuator – Maintenance Practices).

2) Remove actuator rod end and locknut.

3) Remove and retain nut securing transmitter lever. Slacken lever clamp screw and, by turning potentiometer shaft, unscrew lever from potentiometer shaft. Remove lever.

4) Remove the four screws securing bracket and actuator connector or, if same actuator is to be used, remove the two screws securing potentiometer to transmitter bracket. Remove potentiometer.

Note Do not discard attachment items until new potentiometer has been installed on new or replacement actuator (refer to Para 2.B).

B. Install Aileron Trim Tab Position Indicator Potentiometer

Note Coat all bolt shanks with corrosion preventative compound MIL–C–16173 Grade 2 or equivalent before installation.

1) If fitted, remove rod end, rod end locknut and lever nut from new or replacement actuator.

2) Using a 28 Volt DC power supply connected to the appropriate terminals, operate actuator between fully retracted and fully extended, measure overall travel. Operate actuator to the mid-travel position.

3) If actuator is not provided with transmitter attachment bracket, remove the four screws securing actuator connector to actuator body, and install bracket, using bracket and four screws (AN–500A–2–4) retained from actuator previously removed. Tighten and lockwire screws.

4) Install potentiometer to bracket with two screws. Ensure potentiometer shaft is parallel with actuator shaft in both planes. Tighten and lockwire screws.

5) Locate lever on actuator shaft, screwing transmitter potentiometer shaft into lever, until lever contacts actuator shaft shoulder. Install and tighten lever nut, and lockwire to lever.

6) Using a volt/ohm meter, adjust potentiometer shaft (with actuator maintained in the mid-travel position) to give an equal resistance reading between terminals 1 – 2 and 2 – 3. When satisfactory, tighten lever clamp screw.

7) Screw rod end locknut and rod end onto actuator shaft. With actuator fully retracted, adjust rod end until distance between bolt hole centers is 5.19 inches. Tighten rod end locknut.

8) Install trim tab actuator assembly (refer to 27-13-11, Aileron Trim Tab Actuator – Maintenance Practices).

9) Do a test of the aileron trim tab system as per 27-13-00, Aileron Trim Tab System – General Data, Para 4.A.

3. Adjustment/Test

A. Function Test Aileron Trim Tab Position Potentiometer

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Wait several minutes for the aircraft electrical and avionics systems to initialize.

3) Deflect aileron trim to the RW down position (Reference 15 degrees) and note Roll trim display indication correlates and does not become invalid (indicated by an amber ‘X’).

4) Deflect aileron trim to the neutral position (Reference 0 degrees) and note the Roll trim display indication correlates.

5) Deflect aileron trim to the LW down position (Reference 15 degrees) and note the Roll trim display indication correlates and does not become invalid (indicated by an amber ‘X’).

6) If the Roll trim display does not match the aileron trim surface position, repeat the adjustment steps in Para 2..

7) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).